Wind-Assisted Propulsion on North Sea Routes

Maritime Performance Analytics for Flettner Rotors

DataWaves Innovation Laboratory – Preliminary project report

May 2026

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1 Executive summary

The project studies how wind, waves, currents, and vessel heading influence operational speed on North Sea routes, and whether Flettner rotors can improve voyage performance or enlarge useful weather windows. The work follows the advanced variant of the project card: AIS position reports were merged with ERA5 wind fields and Copernicus Marine sea-state/current variables, transformed into trajectory-aware features, modelled with leakage-aware train/test splits, and used in a rotor-assisted what-if simulation.

The final engineered dataset contains 125,160 AIS-metocean observations from 528 voyages between 2023-02-02 and 2026-02-02. The speed target is SOG in knots. The modelling split is chronological by voyage: 422 voyages and 88,346 rows are used for training; 106 later voyages and 35,758 rows are held out for testing. This design prevents observations from the same voyage from appearing on both sides of the split and avoids temporal leakage.

Four model levels were evaluated. A constant-speed baseline gives MAE 1.388 kn and RMSE 1.893 kn. A linear model using wind, wave height, and encoded wind angle gives a small improvement, MAE 1.377 kn and $R^2 = 0.034$. A weather-only XGBoost model is useful for interpretation but does not improve test accuracy under the current feature set. The best predictive model is the lagged nowcast XGBoost model, which adds previous observed SOG and apparent wind context; it reaches MAE 0.406 kn, RMSE 0.758 kn, $R^2 = 0.839$, and 91.7% of predictions within ± 1 kn.

The rotor scenario is deliberately applied after prediction rather than used as a training feature. Rotor power is estimated from the provided polar diagram and converted to a speed increment with the repository assumption 200 kW $\approx$ 1 kn near the operating point. On the held-out test set, the rotor is active in 98.1% of observations, with mean active rotor power 41.2 kW, mean active speed contribution 0.206 kn, and overall mean contribution 0.202 kn. Under the weather-window definition SOG $\geq$ 10 kn and $H_s \leq 3$ m, coverage increases from 76.0% to 77.5%. The number of separate windows decreases from 866 to 808, while points inside windows increase by 529, suggesting that some marginal windows become longer and merge when rotor assistance is applied.

The optional extensions add operational interpretation. The energy model estimates 112.2 t of fuel saved and 349.4 t CO_2 avoided over the three-year dataset, or about 37.4 t fuel and 116.4 t CO_2 per year. Route segmentation shows different benefits by heading regime, with westbound legs gaining the largest weather-window coverage increase. Route optimisation demonstrates a Dijkstra grid-routing proof of concept for rotor gain. Gain detection shows that only 11.9% of observations are high-gain moments, but those moments carry a large share of the total benefit; high gain is associated mainly with wind speeds above 8–12 m/s and beam-to-quartering apparent wind angles.

Table 1: Main quantitative outcomes from the current repository run.

Area	**Current result**
Dataset	125,160 observations, 528 voyages, 2023-02-02 to 2026-02-02.
Validation	Chronological voyage split; 422 train voyages and 106 test voyages; no temporal leakage.
Best speed model	XGBoost lagged nowcast: MAE 0.406 kn, RMSE 0.758 kn, $R^2 = 0.839$.
Rotor speed uplift	Mean active ΔSOG 0.206 kn; overall mean ΔSOG 0.202 kn.
Weather windows	Coverage 76.0% without rotor and 77.5% with rotor; +529 test observations inside windows.
Fuel and emissions	112.2 t fuel saved and 349.4 t CO_2 avoided over the dataset; annualised 37.4 t fuel and 116.4 t CO_2.
High-gain moments	11.9% of observations have ΔSOG $\geq$ 0.5 kn; classifier F1 = 0.992.

2 Project context and objectives

Commercial vessels increasingly use wind-assisted propulsion to reduce fuel burn, emissions, and exposure to volatile fuel prices. A Flettner rotor is a rotating vertical cylinder. When wind flows around it, the Magnus effect creates an aerodynamic lift force. Depending on the apparent wind angle and the vessel's course, part of this force can point forward and support the main engine. In practice the contribution is highly conditional: the same rotor can be useful, neutral, or even unfavourable depending on wind speed, angle, sea state, vessel speed, and operational constraints.

The project card defines the problem as an algorithmic analysis of vessel traces and metocean conditions. The expected outputs are dependencies between environmental conditions and voyage parameters, a scenario comparison between baseline and rotor-assisted operation, and an assessment of weather windows for sailing. The advanced version asks for current/tide vectors, apparent wind calculation, rotor modelling, weather-window estimation, and uncertainty analysis. The repository implements each of these elements.

The practical goal is not to build a final naval-architecture simulator. Instead, the goal is to create a transparent decision-support prototype that can answer questions such as:

- How accurately can SOG be predicted from forecast-available metocean variables and recent vessel behaviour?
- Which environmental variables appear to influence speed under the current data and modelling setup?
- Under what wind-speed and wind-angle combinations do rotors provide meaningful operational gain?
- Do rotors increase the fraction of time in which a voyage satisfies a useful weather-window criterion?
- Can energy, CO_2, route segmentation, and routing analyses be built consistently on top of the same feature set?

3 Data sources and preprocessing

3.1 Raw inputs

The core input is a set of AIS-derived position reports for the analysed vessel. Each point contains time, latitude, longitude, speed over ground, course over ground, and trajectory-derived distance fields. These observations are enriched with gridded metocean data from numerical models:

- ERA5 wind variables: 10 m wind components and gust information.
- Copernicus Marine sea state: significant wave height, mean wave direction, and mean wave period.
- Copernicus Marine currents: eastward and northward current components.

The merged `MasterSet.csv` file contains the base AIS-metocean table. The engineered feature file contains the modelling features used by the baseline and extension notebooks. The feature-engineering notebook reports 125,160 rows and 57 columns after processing.

3.2 Spatio-temporal alignment

AIS points are irregular in time and space, while metocean products are gridded. The preprocessing pipeline aligns these sources by interpolating environmental variables to each AIS position and timestamp. This is the key step that turns a vessel trajectory into a supervised learning table. For each AIS observation the pipeline attaches the environmental state that the vessel experienced at that location and time.

The repository separates this work across proof-of-concept interpolation, ERA5-checking, exploratory-analysis, and feature-engineering notebooks. The resulting dataset spans 2023-02-02 to 2026-02-02 and covers 528 voyages.

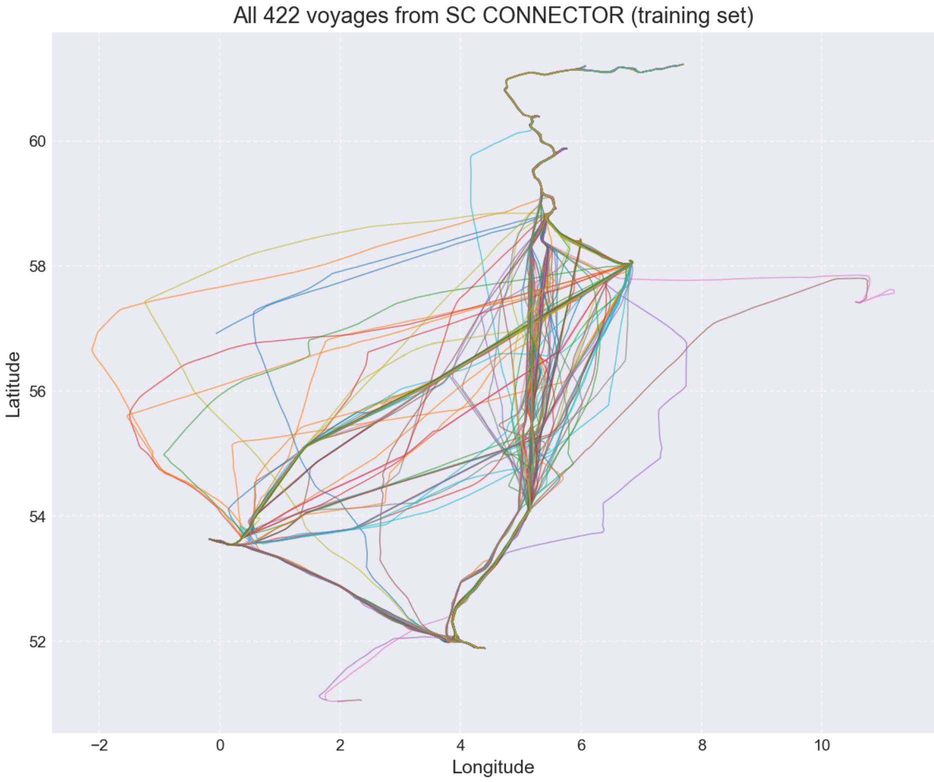


Figure 1: Training trajectories used by the project pipeline.

3.3 Feature engineering

The engineered variables follow a leakage-aware design. Features are divided into explanatory metocean/route variables, lagged nowcast variables, and post-prediction rotor-scenario variables. The main feature groups are:

- **Wind and wave features**: true wind speed, significant wave height H_s, wave period, wind speed squared, and wave-wind alignment.
- **Relative-angle features**: wind direction and angle relative to vessel course, encoded with sine/cosine terms to avoid discontinuity at 0/360 degrees.
- **Apparent wind features**: apparent wind speed and apparent relative angle, reserved for the nowcast model.
- **Current features**: current magnitude, current direction, and along-course/cross-course current components.
- **Lagged target features**: previous SOG, two-step previous SOG, and a shifted rolling mean over up to three previous observations.
- **Interaction terms**: wind-angle, apparent-wind-angle, wind-wave, nonlinear wind/wave terms, and current-course interactions.

The most important leakage control is that the shifted rolling mean never uses the current target row. The lag features are grouped by voyage, so the first observations in a voyage do not inherit speed information from a previous voyage. This makes the nowcast model realistic for short-horizon prediction while keeping the weather-only model interpretable.

Table 2: Representative variables in the engineered feature table.

Group	Examples and role
Trajectory	`Time`, `Lat`, `Lon`, `Voyage_ID`, `Course_deg`, `Dist_since_last_nm`.
Target	`Speed_kn`; the predicted speed over ground.
Wind/wave	`True_Wind_Speed_ms`, `H_s`, `mwp`, `fg10`, wave-wind alignment.
Angles	`alpha_sin`, `alpha_cos`, apparent wind angle encodings.
Currents	`Current_Speed_ms`, along-course and cross-course current terms.
Nowcast history	`Speed_kn_lag1`, `Speed_kn_lag2`, `Speed_kn_mean3`.
Rotor scenario	Rotor power, active flag, and ΔSOG are used after model prediction, not as training features.

4 Methodology

4.1 Speed prediction problem

The primary supervised learning target is vessel speed over ground:

$$\widehat{SOG}_t = f(X_t),$$

where X_t contains environmental variables, route context, and, for the nowcast model only, previous observed speed. The models are evaluated on later voyages that were not used for training. This is essential because random row splits would allow neighbouring points from the same voyage to appear in both train and test sets, giving overly optimistic performance.

The project uses three conceptual model families:

1. **B1 constant mean**: predicts the mean training speed for every test observation.
2. **B2 linear weather model**: predicts speed from true wind speed, H_s, and sine/cosine encoding of the relative wind angle.
3. **XGBoost models**: gradient-boosted trees trained in two modes: weather-only explanatory modelling and lagged nowcast prediction.

The XGBoost implementation uses 300 estimators, maximum depth 6, learning rate 0.05, 80% row subsampling, 80% column subsampling, and a fixed random seed. Features are standardised before model fitting in the current notebook implementation.

4.2 Weather-only versus nowcast modelling

The report distinguishes two purposes that are easy to confuse:

- The **weather-only model** is an explanatory model. It excludes target-history features, so its behaviour is more directly related to wind, waves, currents, and route context.
- The **lagged nowcast model** is a predictive operational model. It uses previous speed observations and apparent wind context, making it much more accurate for short-horizon prediction.

This separation matters for interpretation. A high-performing nowcast model can mostly learn persistence: ships tend to continue moving near their recent speed. That is useful operationally, but it does not by itself prove that weather features are strongly explanatory. The repository therefore reports both model types and treats weather interpretation with caution.

4.3 Rotor scenario model

The rotor scenario is applied after baseline speed prediction. For each AIS observation, the rotor utility estimates a propulsion power P_{rotor} from wind speed and relative wind angle using the supplied polar-response table. The rotor is active if:

$$P_{\text{rotor}} > 0, \quad H_s \leq 6 \text{ m}, \quad U_{\text{wind}} \leq 42 \text{ m/s}.$$

The speed contribution is then approximated as:

$$\Delta SOG = \frac{P_{\text{rotor}}}{200\text{kW/kn}}$$

when the rotor is active, otherwise $\Delta SOG = 0$. This is a transparent surrogate assumption rather than a full hydrodynamic propulsion model. Its value is that it makes the scenario auditable: the rotor benefit can be traced back to the wind-speed/angle polar, wave-height gate, and a single power-to-speed conversion.

4.4 Weather-window definition

The main weather-window criterion follows the project card:

$$SOG \geq 10\text{kn} \quad \text{and} \quad H_s \leq 3 \text{ m}.$$

For every test-set observation the pipeline compares a baseline prediction against a rotor-assisted prediction. Contiguous observations satisfying the criterion form weather windows. The report tracks both the number of windows and the number/share of observations inside windows. Counting both is important: the rotor can reduce the number of separate windows if neighbouring windows merge into longer continuous periods.

4.5 Uncertainty evaluation

The baseline notebook also trains quantile-regression prediction intervals for the nowcast speed model. The reported 95% interval has mean lower bound 9.670 kn, mean upper bound 12.831 kn, mean width 3.161 kn, and empirical coverage 96.5% on the test set. This suggests that the interval model is slightly conservative but well calibrated for the current test distribution.

5 Model results and evaluation

5.1 Dataset split

The train/test split is chronological by voyage. The first 80% of voyages by start time are used for training; the final 20% are used for testing. The baseline notebook reports:

- Training set: 88,346 rows from 422 voyages.
- Test set: 35,758 rows from 106 voyages.
- Training period: 2023-02-02 to 2025-05-23.
- Test period: 2025-05-23 to 2026-02-02.
- Temporal leakage check: true.

This split is stricter than a random split and is therefore a stronger estimate of generalisation to future voyages.

5.2 Regression metrics

The model comparison is summarised in Table 3. The constant mean and linear weather model are close to one another. The weather-only XGBoost model performs poorly relative to the linear baseline on the held-out test period, which suggests either distribution shift, missing operational variables, or that a large part of speed variation is driven by control decisions and vessel state not visible in the current feature table. The lagged nowcast model is much stronger because recent observed speed is a powerful predictor.

Table 3: Held-out test performance for the core model hierarchy.

Model	**MAE** [kn]	**RMSE** [kn]	R^2	**Within** ± 1 kn
B1: Constant mean	1.388	1.893	-0.002	49.3%
B2: Linear wind/wave/angle	1.377	1.860	0.034	49.5%
M1: XGBoost weather-only	1.468	1.883	0.010	41.7%
M2: XGBoost lagged nowcast	0.406	0.758	0.839	91.7%

Figure 2: Predicted versus actual SOG for the lagged nowcast model.

5.3 Feature importance

The weather-only model ranks course and current interaction terms highly, while the lagged nowcast model is dominated by recent observed speed. This is consistent with vessel operations: the current speed carries information about engine setting, traffic, manoeuvring, loading, and other operational factors that are not explicitly modelled.

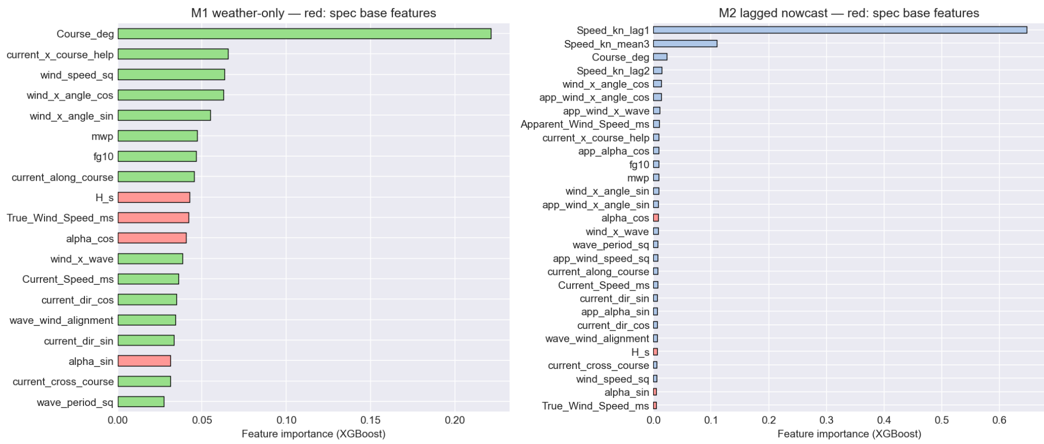


Figure 3: Feature importances for weather-only and lagged nowcast XGBoost models.

The main interpretation is therefore twofold. First, the project can now predict short-term SOG accurately when recent speed is available. Second, pure weather-to-speed inference is harder than expected and should be treated as an explanatory sensitivity tool rather than a final operational predictor.

6 Rotor and weather-window results

6.1 Rotor what-if scenario

On the test set, the rotor surrogate is active for 35,077 of 35,758 rows, or 98.1% of observations. The mean rotor power when active is 41.2 kW. The mean active speed increment is 0.206 kn, and the overall mean increment is 0.202 kn. Per-voyage results are heterogeneous: the highest-gain voyages in the test set show mean speed contributions from about 0.47 kn to 1.19 kn, while the overall mean remains modest because many observations have weaker wind, less favourable angle, or low rotor power.

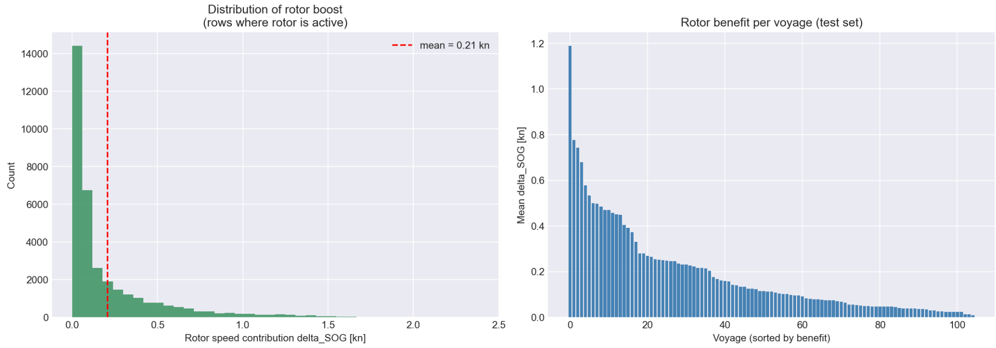


Figure 4: Rotor scenario distribution and per-voyage mean speed contribution.

The result is operationally meaningful but not spectacular in average speed terms. An average 0.2 kn speed uplift can still matter over long voyages, especially if it moves marginal periods over a speed threshold or allows lower engine load for the same schedule. The larger gains occur in a smaller subset of conditions, which motivates the gain-detection extension.

6.2 Weather-window impact

The weather-window analysis evaluates whether rotor assistance helps maintain the criterion SOG ≥ 10 kn and $H_s \leq 3$ m. The result is:

Table 4: Weather-window comparison on the held-out test set.

Scenario	Windows	Points in window	Coverage
Without rotor	866	27,167	76.0%
With rotor	808	27,696	77.5%
Delta	-58	+529	+1.5 percentage points

The decrease in the number of windows should not be read as a negative result. When rotor assistance lifts speed above the threshold between two favourable periods, separate windows can merge. The more relevant operational result is that the number of points inside the window increases and total coverage rises.

6.3 High-gain geography and environmental structure

The gain-surface and interaction plots show that rotor benefits cluster in particular wind-speed and angle regimes. This is expected from the polar diagram: headwind and tailwind directions are poor, while beam and broad-reaching angles are stronger. Wave height mostly acts as a boundary or risk condition; in the implemented scenario the rotor is forced off above $H_s = 6$ m, and only 19 observations exceed that threshold in the full engineered dataset.

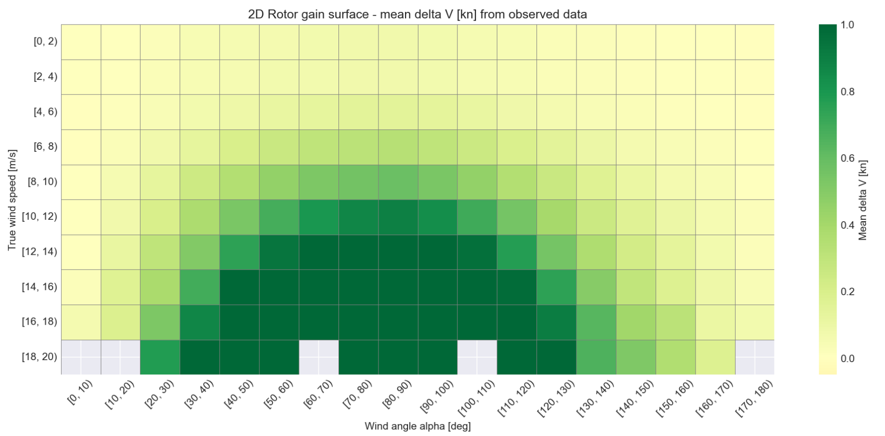


Figure 5: Environmental structure of the rotor-gain scenario.

7 Energy, emissions, and CII extension

The first extension converts rotor power into an energy and emissions impact. The ship model assumes 3,500 kW at design speed 11.5 kn, a cubic resistance relationship, specific fuel oil consumption of 195 g/kWh, and an HFO CO_2 factor of 3.114 t CO_2/t fuel. Under these assumptions the fuel rate at design speed without the rotor is 16.38 t/day.

The three-year dataset summary is:

- Total observations: 125,160.
- Rotor active share: 97.6%.
- Mean brake power: 3,887 kW.
- Mean rotor power when active: 40.4 kW.
- Total fuel baseline: 11,066.7 t.
- Total fuel with rotor: 10,954.4 t.
- Fuel saved: 112.2 t, or 1.01%.
- CO_2 saved: 349.4 t, or 1.01%.

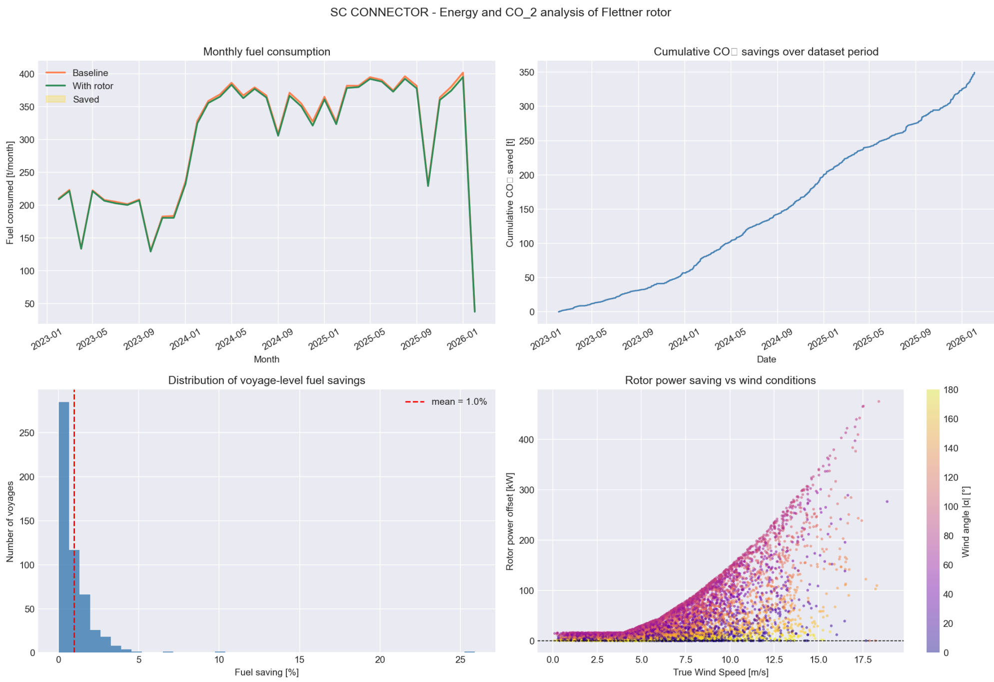


Figure 6: Fuel and CO_2 impact estimated from the rotor energy model.

Annualised over the dataset duration of 1,096 days, the extension estimates 37.4 t fuel saved per year and 116.4 t CO_2 avoided per year. The attained CII estimate improves from 53.75 to 53.20 g CO_2/(t nm), a 1.01% improvement.

The conclusion is that the average annual energy impact is modest for one rotor, but it is measurable and directionally useful. The result should be interpreted as an order-of-magnitude scenario rather than a guaranteed vessel-performance claim, because the model does not yet include detailed propeller curves, engine load-specific SFOC, hull fouling, loading condition, RPM control policy, or captain/charterer speed decisions.

8 Route segmentation extension

Route segmentation divides the data into heading regimes: northbound, eastbound, southbound, and westbound. This makes the analysis more interpretable because rotor performance is strongly tied to the relationship between course and wind. A single global model can hide differences between legs moving through different prevailing-wind relationships.

Table 5: Regime-level characterisation from the full dataset.

Regime	**Mean SOG** [kn]	**Mean wind** [m/s]	**Mean H_s** [m]	**Mean ΔSOG** [kn]
Northbound	12.05	7.0	1.32	0.209
Eastbound	12.34	6.2	1.06	0.185
Southbound	10.95	6.8	1.28	0.196
Westbound	10.58	6.5	1.21	0.209

The per-regime weather-only XGBoost models show mixed predictive quality. Northbound has the lowest MAE at 1.219 kn, while westbound is hardest to predict at MAE 1.788 kn. Eastbound is the only regime with clearly positive R^2 in the per-regime model summary, $R^2 = 0.137$, while the other regimes have negative R^2. This again supports the conclusion that weather-only generalisation is difficult.

The most useful output from this extension is the weather-window effect by heading. Westbound legs improve from 85.1% to 90.1% coverage, the largest regime-level gain. Southbound improves from 82.3% to 84.5%. Northbound and eastbound already have high baseline coverage and therefore show smaller coverage changes.

9 Route optimisation extension

The route-optimisation extension explores whether route choice can increase the expected rotor benefit. The implemented proof of concept builds a coarse 0.5-degree spatial grid from historical metocean/rotor-gain fields and applies Dijkstra search to identify a path with improved integrated rotor gain. The reference voyage is Voyage 456, from approximately 51.89 degrees N, 4.40 degrees E to 61.23 degrees N, 7.68 degrees E. The great-circle distance is about 570.7 nm.

The spatial field contains 72 grid cells with data. Mean rotor power across the grid is 78.5 kW, and the best historical cells reach mean rotor powers above 180–290 kW under favourable conditions. The route-search result reports an optimal path of 19 nodes, distance 568.0 nm versus 571.0 nm for the direct grid path, mean rotor power 7.5 kW versus 1.9 kW direct, and mean ΔSOG 0.037 kn versus 0.010 kn direct. Relative improvement is large at +288.5%, but the absolute gain in this specific route experiment is small.

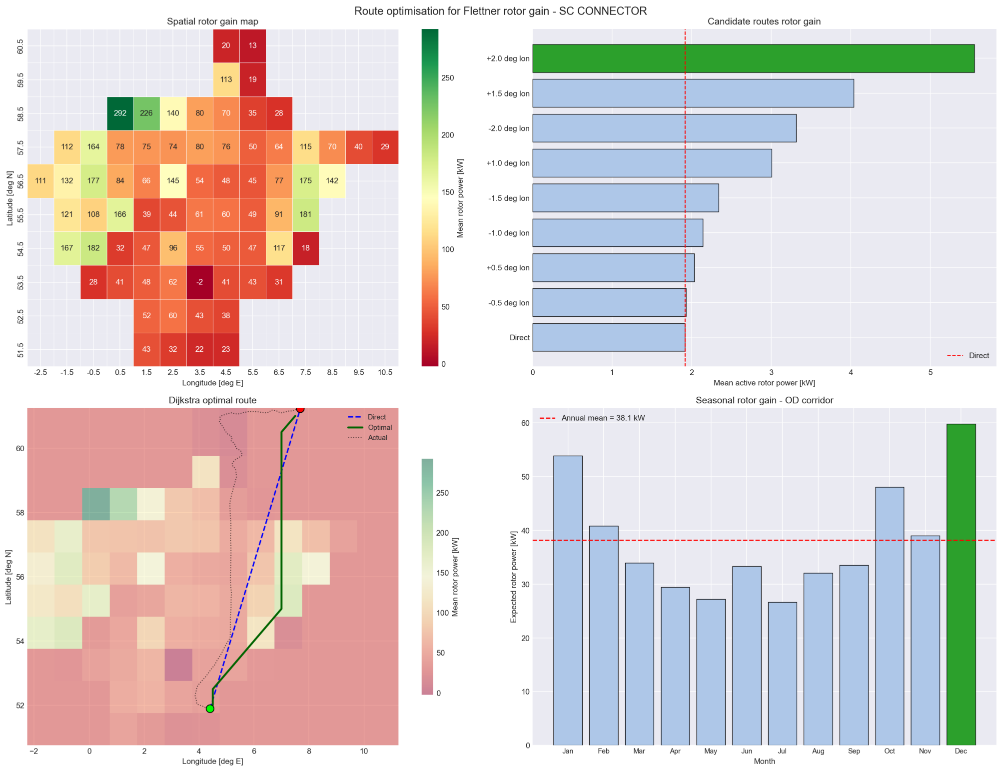


Figure 7: Dijkstra-based routing proof of concept for rotor-gain optimisation.

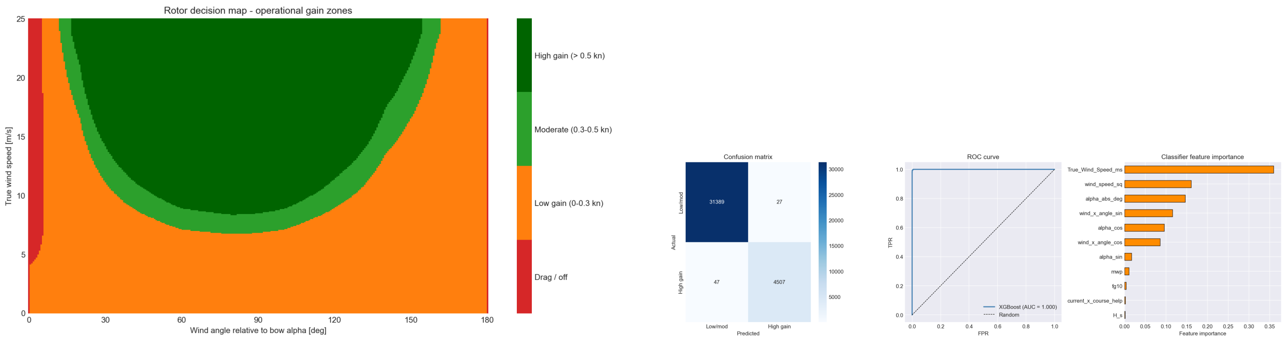
The extension also studies seasonal sensitivity. December has the best expected monthly rotor power, 59.8 kW, while July is worst at 26.6 kW. The seasonal pattern is intuitive for the North Sea: winter has stronger winds, but it may also have more severe waves and operational risk. A final routing product should therefore optimise a multi-objective score, not rotor gain alone. A practical score would combine expected arrival time, fuel, CO_2, wave limits, route deviation, safety margin, and schedule reliability.

10 Rotor gain detection extension

The gain-detection extension turns the rotor scenario into a classification and decision problem. A high-gain observation is defined as $\Delta SOG \geq 0.5$ kn. Under the current surrogate model, high-gain moments account for 14,856 observations, or 11.9% of the full dataset.

Rotor gain increases strongly with wind speed. The notebook reports mean ΔSOG of 0.037 kn for wind below 4 m/s, 0.273 kn for 8–10 m/s, 0.431 kn for 10–12 m/s, 0.610 kn for 12–14 m/s, and 0.897 kn above 14 m/s. The optimal high-gain subset has median wind speed 11.5 m/s, and 90% of those observations are above 9.2 m/s.

Angle is equally important. High-gain observations have median absolute relative wind angle around 82 degrees, with 80% of the main high-gain subset between 50 and 113 degrees. This matches the rotor polar: beam and quartering winds are useful, while headwind and tailwind sectors are poor.



(a) Rule-based high-gain zones. **(b)** Classifier performance.

Figure 8: Operational detection of high-gain rotor conditions.

An XGBoost classifier identifies high-gain moments with 99.8% accuracy, 99.4% precision, 99.0% recall, F1 = 0.992, and ROC-AUC = 1.000 on the notebook split. These very high metrics are plausible because the target is derived directly from the same rotor-power surrogate and its input variables; the classifier is therefore best interpreted as a compact decision-rule learner, not as independent proof of physical rotor performance.

The rule-based decision system is more transparent: if wind is at least 8 m/s and angle is between 50 and 130 degrees, or wind is at least 12 m/s and angle is between 40 and 140 degrees, mark high gain. This rule reaches 95.8% recall, 69.7% precision, and 94.2% accuracy. For operations, the rule is attractive because it is easy to explain and can be applied to forecasts.

11 Conclusions

The project has progressed beyond a simple baseline and now forms a coherent maritime analytics prototype. AIS trajectories are enriched with wind, waves, and currents; leakage-aware models are trained and evaluated by voyage; rotor assistance is simulated as a transparent what-if layer; and the results are translated into weather windows, fuel, emissions, route regimes, routing, and high-gain operational rules.

The central technical finding is that recent vessel speed is the strongest predictor of near-term speed. The lagged nowcast model reaches MAE 0.406 kn and $R^2 = 0.839$, while weather-only models remain much weaker. This distinction should be presented honestly: the project can predict speed well in a nowcast setting, but causal weather-only interpretation is still limited by missing operational variables.

The central rotor finding is that the average benefit of one rotor is moderate but operationally relevant. Mean speed uplift is about 0.2 kn across the test set, weather-window coverage increases by 1.5 percentage points, and the energy extension estimates about 37 t/year fuel saving and 116 t/year CO_2 reduction. The most useful message is conditionality: the rotor is most valuable in a minority of high-gain moments, especially when wind is above roughly 8–12 m/s and the apparent wind angle is near beam/quartering sectors.

12 Transparency statement

This report was partly done with the assistance of generative AI tools. Mostly the paraphrasing and summarisation. The AI was used to help condense and clarify the technical content, but all key findings, interpretations, and conclusions were generated by the authors. The AI did not have access to the raw data or code and did not make any modelling decisions. The report was reviewed and edited by the authors to ensure accuracy and relevance.